

(a) Twin construction



A: dog B: rooster

Reverse query $\leftrightarrow$ **swap target**

q_+ : dog followed by rooster $\rightarrow W_+$

q_- : rooster followed by dog $\rightarrow W_-$

(b) RelTwin training

Stage 1: Answer learning

	$y(W_+)$	$y(W_-)$
q_+	s_{++} $\uparrow$	s_{+-} $\downarrow$
q_-	s_{-+} $\downarrow$	s_{--} $\uparrow$

Competition + reversal consistency

Stage 2: Set preferences



Inference Audio + query $\rightarrow$ Adapted backbone $\rightarrow$ Timestamps

(c) Window selection

